

Training strategy	MATH500	AMC23	AIME2024	AIME2025	BBEH	Average
GRPO (no denoise prefix)	83.6	63.1	22.1	18.1	11.1	39.6
DenoiseRL: small weak model	85.8	61.4	24.8	23.3	14.8	42.0
DenoiseRL: frozen pre-RL self	86.2	65.2	21.0	22.4	12.2	41.4
DenoiseRL: random-token noise	84.6	63.8	20.8	20.8	11.5	40.3